Candidate Realistic 3D Problems for the Dynamic OT / Schrödinger-Bridge Solver

Optimal Transport / Schrödinger Bridge Project
Companion to the Python/JAX 3D code in `python/3d`

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Abstract

This note collects verification problems with reference solutions and realistic application problems that the 3D solver could tackle for the paper. It is a literature and data survey: nothing here has been run. Items that could not be verified from the sources are flagged. The solver's constraints, as assumed throughout, are: a uniform Cartesian grid on $[0,1]^3$, no-flux (Neumann) walls, nonnegative marginals of equal mass, and $n_t = n_x = n_y = n_z$ up to 96 in double precision (112 in the `mixedc` mode) on one GH200. Section 4 gives a ranked shortlist.

1 A convention warning that affects every closed form

All the Gaussian closed forms below are stated for the reference SDE $dX_t = \sigma\, dW_t$ on $[0,1]$, equivalently for the static problem

$$\min_{\pi} \int |x-y|^2\, d\pi \;+\; 2\sigma^2\, \mathrm{KL}(\pi \,|\, \alpha \otimes \beta)$$

(Janati et al.). The Fokker–Planck equation of that bridge is

$$\partial_t \rho + \nabla \cdot (\rho v) = \tfrac{\sigma^2}{2}\, \Delta \rho.$$

If our diffusion term is written $\varepsilon\, \Delta\rho$ then $\sigma^2 = 2\varepsilon$; if it is written $\frac{\varepsilon}{2}\Delta\rho$ then $\sigma^2 = \varepsilon$. Check this before comparing costs or ρ_t against any formula in Section 2.

2 Verification problems with reference solutions

2.1 Anisotropic / unequal-covariance Gaussians (closed form)

This is the top verification pick: cost, ρ_t and m are all in closed form, and it exercises the cross-axis coupling that the isotropic test cannot see.

Bridge marginals. Bunne, Hsieh, Cuturi, Krause, "The Schrödinger Bridge between Gaussian Measures has a Closed Form", AISTATS 2023 (PMLR 206:5802–5833), arXiv:2202.05722, Theorem 3 (constant σ, Brownian reference). With $\Sigma = \Sigma_0$, $\Sigma' = \Sigma_1$,

$$D_\sigma = \big(4\,\Sigma^{1/2}\Sigma'\Sigma^{1/2} + \sigma^4 I\big)^{1/2}, \qquad C_\sigma = \tfrac{1}{2}\big(\Sigma^{1/2} D_\sigma \Sigma^{-1/2} - \sigma^2 I\big), \tag{1}$$

$$\mu_t = (1-t)\mu_0 + t\mu_1, \tag{2}$$

$$\Sigma_t = (1-t)^2\Sigma_0 + t^2\Sigma_1 + t(1-t)\big(C_\sigma + C_\sigma^\top + \sigma^2 I\big), \tag{3}$$

(their eqs. 26–27; C_σ satisfies $C_\sigma^2 + \sigma^2 C_\sigma = \Sigma\Sigma'$, eq. B.18). The optimal SDE is $dX = f(t,X)\,dt + \sigma\,dW$ with $f(t,x) = S_t^\top \Sigma_t^{-1}(x-\mu_t) + \dot\mu_t$ (eqs. 28–29); take S_t from the PDF, the ar5iv rendering is garbled.

Cost. Janati, Muzellec, Peyré, Cuturi, "Entropic optimal transport between (unbalanced) Gaussian measures has a closed form", NeurIPS 2020, arXiv:2006.02572, Theorem 1 (balanced case): the optimal plan is $\pi^* \sim \mathcal{N}\big((a,b), \left[\begin{smallmatrix} A & C_\sigma \\ C_\sigma^\top & B \end{smallmatrix}\right]\big)$ with the same D_σ, C_σ, and

$$\mathrm{OT}_\sigma = \|a-b\|^2 + \operatorname{Tr} A + \operatorname{Tr} B - \operatorname{Tr} D_\sigma + d\,\sigma^2\big(1 - \log 2\sigma^2\big) + \sigma^2 \log\det\big(D_\sigma + \sigma^2 I\big).$$

Our objective may omit the entropy term: compare the kinetic-energy part via the plan, or add the KL term. Mallasto, Gerolin, Minh, "Entropy-regularized 2-Wasserstein distance between Gaussian measures", Information Geometry 5 (2022) 289–323, arXiv:2006.03416, is an independent derivation of the same quantities, useful for cross-checking sign and normalization conventions.

Momentum without the drift formula. The SB current velocity is a gradient, and for Gaussian marginals it is affine: $v_t(x) = \dot\mu_t + M_t(x-\mu_t)$ with symmetric M_t solving the Lyapunov equation $M_t\Sigma_t + \Sigma_t M_t = \dot\Sigma_t$ (from the continuity equation; $\dot\Sigma_t$ is explicit from (3)). Then $m(t,x) = \rho_t(x)\,v_t(x)$. This is the survey's own derivation: verify it against eq. 29 minus $\frac{\sigma^2}{2}\nabla\log\rho_t$.

Fit and sub-cases. Perfect for the box as long as 3–4 standard deviations of every Σ_t stay inside $[0,1]^3$; note that Σ_t bulges above the linear interpolant by $t(1-t)\sigma^2 I$. Good sub-cases:

- rotated anisotropic covariance, $\Sigma_1 = R\,\Sigma_0 R^\top$;
- $\Sigma_0 = \Sigma_1$ with $\mu_0 \neq \mu_1$ (translation with an ε-dependent bulge);
- the "self-bridge" $\rho_0 = \rho_1$: classically static, but the SB marginal covariance is $\Sigma_t = \Sigma + t(1-t)\big(C_\sigma + C_\sigma^\top + \sigma^2 I - 2\Sigma\big)$, a pure-$\varepsilon$ effect.

Effort: low. Paper value: essential; reviewers will expect it.

2.2 Separable (product) problems: a free reference from the 1D code

For quadratic cost the Gibbs kernel factorizes, so the bridge between product measures $\rho_0 = \bigotimes_i a_i$, $\rho_1 = \bigotimes_i b_i$ is the product of the three 1D bridges (uniqueness of the entropic plan). This also holds on the Neumann box, because the Neumann heat kernel factorizes. Any non-Gaussian 1D pair solved to high accuracy in 1D (bimodal → unimodal, box → bump, ...) therefore gives an exact 3D $\rho(t,x)$ and m. This is the 2D product-Sinkhorn trick one dimension up. Effort: trivial. Paper value: moderate, but a strong code check, especially of anisotropy handling.

2.3 Grid-exact Sinkhorn reference on the Neumann box (any pair)

The SB time marginals are $\rho_t = (P_t\varphi)(P_{1-t}\psi)$ with P the reference heat semigroup and φ, ψ the Schrödinger potentials (Léonard's survey). Running Sinkhorn on our own grid with the Neumann heat kernel applied by DCT gives the exact discrete-space SB for the same no-flux boundary condition, for any ρ_0, ρ_1 (this is the 2D product-Sinkhorn reference extended to 3D with cosine transforms). A few hundred DCT-Sinkhorn iterations at 96^3 is cheap. It works for every application in Section 3, gives ρ_t (not m) at all t, and has no time discretization, so it isolates the LADMM / time-stepping error.

2.4 Constructed pairs with known SB (Gushchin et al.)

Gushchin, Kolesov, Mokrov, et al., "Building the Bridge of Schrödinger: A Continuous Entropic Optimal Transport Benchmark", NeurIPS 2023, arXiv:2306.10161, code `https://github.com/ngushchin/EntropicOTBenchmark`. The paper is written for neural EOT solvers that only ever see *samples*, which makes its language foreign to a grid code; this subsection translates it, then turns it into a closed-form multimodal test with explicit ρ_0, ρ_1, ρ_t and m. The formulas are implemented (numpy, any dimension) in `python/shared/mixture_sb.py`; nothing has been run through the solver.

Convention. The paper's EOT problem is $\min_\pi \int \frac{1}{2}|x-y|^2\, d\pi + \varepsilon_{\mathrm{p}}\, \mathrm{KL}(\pi \,|\, \rho_0 \otimes \rho_1)$ with reference SDE $dX = \sqrt{\varepsilon_{\mathrm{p}}}\, dW$. Our constraint is $\partial_t \rho + \nabla \cdot m = \varepsilon \Delta \rho$, i.e. $\sigma^2 = 2\varepsilon$, so

$$\varepsilon_{\mathrm{p}} = \sigma^2 = 2\varepsilon \qquad \text{(their } \varepsilon \text{ is twice ours).}$$

Everything below is written with $\sigma^2 = 2\varepsilon$ so it can be pasted against the code.

The one idea: pick the answer first. The optimal entropic plan is characterized by its Gibbs form. A coupling π of ρ_0, ρ_1 is optimal if and only if

$$\pi^*(x,y) = \hat{\varphi}_0(x)\, \exp\!\Big(-\tfrac{|x-y|^2}{2\sigma^2}\Big)\, \varphi_1(y) \tag{4}$$

for some positive functions $\hat{\varphi}_0, \varphi_1$ (the Schrödinger factors; in potential language $\hat{\varphi}_0 = e^{f^*/\sigma^2}$, $\varphi_1 = e^{g^*/\sigma^2}$). Read backwards: choose *any* positive pair $(\hat{\varphi}_0, \varphi_1)$, define π^* by (4), and π^* is automatically the optimal plan between its own two marginals. A "constructed pair" is a (ρ_0, ρ_1) obtained this way, so the exact solution is known before the problem is posed.

What the paper does, and what "sampling" means. The paper fixes ρ_0 (a Gaussian, or a normalizing flow) and fixes the potential on the y side, g^* (their f^*). Their Theorem 3.2 is the conditional of (4):

$$\pi^*(y \,|\, x) = \frac{\exp\big((g^*(y) - \frac{1}{2}|x-y|^2)/\sigma^2\big)}{Z(x)}, \qquad Z(x) = \int \exp\big((g^*(y) - \tfrac{1}{2}|x-y|^2)/\sigma^2\big)\, dy, \tag{5}$$

so $\hat{\varphi}_0 = \rho_0 / Z$ is whatever it has to be for the x-marginal to be ρ_0. Their eq. (15) takes g^* log-sum-exp quadratic, $e^{g^*(y)/\sigma^2} = \sum_n w_n\, Q(y \,|\, b_n, \sigma^{-2} A_n)$ with $Q(y|b,A) = e^{-\frac{1}{2}(y-b)^\top A (y-b)}$, which is (up to the weights) a Gaussian mixture $\sum_n w_n\, \mathcal{N}(y \,|\, b_n,\, \sigma^2 A_n^{-1})$. Multiplying a Gaussian in

y by the Gaussian kernel $e^{-|x-y|^2/2\sigma^2}$ gives a Gaussian in y, so Prop. 3.3 follows: $\pi^*(\cdot|x)$ is a Gaussian mixture with

$$\text{means } (A_n{+}I)^{-1}(A_n b_n{+}x), \qquad \text{covariances } \sigma^2(A_n{+}I)^{-1}, \qquad \text{weights } \propto w_n\,\mathcal{N}(x\,|\,b_n,\,\sigma^2(A_n^{-1}{+}I)).$$

That is all the "sampler" is: draw $x\sim\rho_0$, pick a component n with the x-dependent weights, draw y from that Gaussian. The pair (x,y) is a sample of the optimal plan; throwing x away gives a sample of $\rho_1(y)=\int\rho_0(x)\pi^*(y|x)\,dx$. The paper never writes ρ_1 as a density because its consumers never need one: a neural solver is trained from batches of x's and y's. A grid solver needs the density, and $\rho_1=\varphi_1\cdot(\text{heat flow of }\rho_0/Z)$ is a D-dimensional Gaussian convolution of a known function: cheap in 2D by quadrature, FFT-able in 3D, but an extra numerical step whose error would have to be tracked. The variant below removes it.

Closed-form variant: both factors Gaussian mixtures. Choose both endpoint factors as positive Gaussian mixtures (positive definite P_i, Q_n; this is the sub-case $A_n\succ 0$ of the paper's family),

$$\hat\varphi_0(x)=\sum_{i=1}^{I}u_i\,\mathcal{N}(x\,|\,a_i,P_i),\qquad \varphi_1(y)=\sum_{n=1}^{N}w_n\,\mathcal{N}(y\,|\,b_n,Q_n).$$

The factors evolve by the heat equation with generator $\varepsilon\Delta$ ($\partial_t\varphi=-\varepsilon\Delta\varphi$ backward from φ_1, $\partial_t\hat\varphi=\varepsilon\Delta\hat\varphi$ forward from $\hat\varphi_0$), which just adds $2\varepsilon t$ to every covariance:

$$\varphi(t,z)=\sum_n w_n\,\mathcal{N}\big(z\,|\,b_n,\,Q_n+2\varepsilon(1-t)I\big),\qquad \hat\varphi(t,z)=\sum_i u_i\,\mathcal{N}\big(z\,|\,a_i,\,P_i+2\varepsilon t\,I\big),$$

and $\rho_t=\varphi\,\hat\varphi/K$. Each product of two Gaussians is a Gaussian times a constant, so

$$\rho_t(z)=\sum_{i,n}\omega_{in}\,\mathcal{N}\big(z\,|\,c_{in}(t),\,C_{in}(t)\big),\qquad \omega_{in}=\frac{u_i w_n\,\mathcal{N}\big(a_i\,|\,b_n,\,P_i+Q_n+2\varepsilon I\big)}{K},\tag{6}$$

$$C_{in}(t)=\Big[(P_i+2\varepsilon tI)^{-1}+(Q_n+2\varepsilon(1-t)I)^{-1}\Big]^{-1},\qquad c_{in}(t)=C_{in}(t)\Big[(P_i+2\varepsilon tI)^{-1}a_i+(Q_n+2\varepsilon(1-t)I)^{-1}b_n\Big]$$

with K the sum of the numerators of ω_{in} (so $\sum\omega_{in}=1$). ρ_0 and ρ_1 are (6) at $t=0,1$. The branch weights ω_{in} do not depend on t: ω_{in} is exactly the mass that travels from source blob i to target blob n, so the static plan is readable by eye. The optimal drift and momentum are

$$v(t,z)=\sigma^2\nabla\log\varphi(t,z)=2\varepsilon\,\frac{\sum_n w_n\,\mathcal{N}(z|b_n,S_n(t))\,\big(-S_n(t)^{-1}(z-b_n)\big)}{\sum_n w_n\,\mathcal{N}(z|b_n,S_n(t))},\quad S_n(t)=Q_n{+}2\varepsilon(1{-}t)I,\qquad m=\rho_t v.\tag{7}$$

Note v is *not* the ω-weighted average of the per-branch Gaussian velocities; the branches interact through the denominator. Checks done (`shared/mixture_sb.py`, numpy, 2D and 3D): (6) equals $\varphi\hat\varphi/K$ to 10^{-15}; $\int\rho_t=1$; ρ_1 from (6) equals $\int\rho_0\,\pi^*(y|x)\,dx$ by quadrature to $3\cdot10^{-13}$; the finite-difference residual of $\partial_t\rho+\nabla\cdot(\rho v)-\varepsilon\Delta\rho$ falls as $2.3\cdot10^{-3}, 5.7\cdot10^{-4}, 1.4\cdot10^{-4}, 3.6\cdot10^{-5}$ on $200^2,\dots,1600^2$ grids, i.e. exactly $O(h^2)$; with $I=N=1$ and $P=Q$ the centres, variances and velocity coincide with `analytical_sb_gaussian_2d` (the $\alpha=\sqrt{\sigma_g^4+\varepsilon^2}-\sigma_g^2$ formula) to all printed digits. The kinetic energy $\iint|m|^2/2\rho$ has no closed form; it is a cheap quadrature of (6)–(7) on a fine grid.

Worked 2D example (one blob splits into two). $\varepsilon = 0.01$, $\hat{\varphi}_0 = \mathcal{N}((0.30, 0.50),\, 0.05^2 I)$, $\varphi_1 = \frac{1}{2}\mathcal{N}((0.70, 0.30),\, 0.04^2 I) + \frac{1}{2}\mathcal{N}((0.70, 0.70),\, 0.04^2 I)$. The resulting marginals are (`example_split_2d`):

	branch mass ω	centre at $t = 0$	centre at $t = 1$	std ($t = 0$ / $t = 1$)
branch $n = 1$	0.5	$(0.342, 0.479)$	$(0.673, 0.313)$	0.047 / 0.039
branch $n = 2$	0.5	$(0.342, 0.521)$	$(0.673, 0.687)$	0.047 / 0.039

At $t = 0$ the two branches overlap into one slightly elongated blob; at $t = 1$ they are two separate blobs. Note how ρ_0 is *not* $\hat{\varphi}_0$: the centre is pulled from 0.30 to 0.342 by $\varphi(0, \cdot)$, and the width shrinks from 0.05 to 0.047. One does not get to prescribe ρ_0 and ρ_1 directly; one prescribes the factors and reads the marginals off (6), then nudges $a_i, b_n, P_i, Q_n, u_i, w_n$ until the marginals look as wanted. Mass of ρ_t inside $[0, 1]^2$ is $\geq 1 - 2 \cdot 10^{-7}$ for all t; kinetic energy 0.0922. The problem is not a product and not Gaussian, so it tests the cross-axis coupling and the nonlinearity of the prox in a way Sec. 2.1 cannot (Fig. 1). When the time comes to compare, the reference to compare against is (6) for ρ and (7) for m, sampled on the solver's staggered grid exactly as `analytical_sb_gaussian_2d` does.

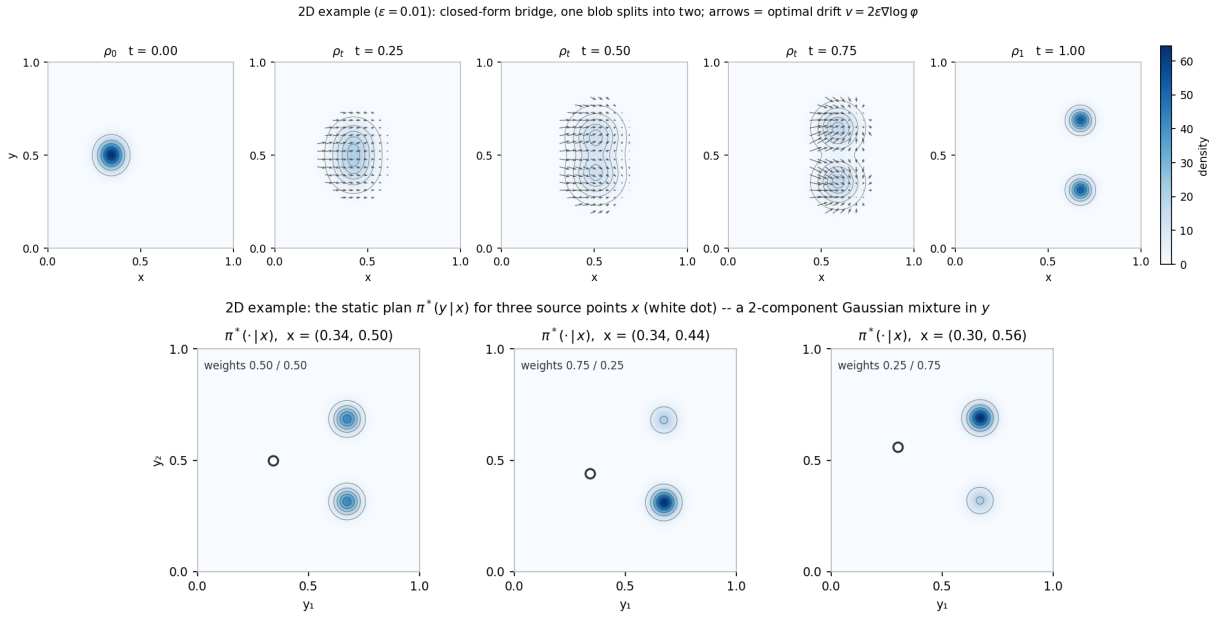


Figure 1: 2D example, all from the closed form (no solver run). Top: ρ_0, three intermediate ρ_t with the optimal drift (7) as arrows, and ρ_1; shared colour scale, contours at 10–90% of each panel's own peak. The bridge spreads to roughly twice its endpoint width at $t = \frac{1}{2}$ (the ε bulge of Sec. 2.1) and recondenses. Bottom: the static plan $\pi^*(y \,|\, x)$ for three source points: a two-component Gaussian mixture in y whose weights tilt toward the nearer target (this is the paper's Prop. 3.3 object, and what its "sampler" draws from).

Worked 3D example (two blobs go to three). $\varepsilon = 0.005$, $\hat{\varphi}_0 = \frac{1}{2}\mathcal{N}((0.30, 0.35, 0.50), 0.05^2 I) + \frac{1}{2}\mathcal{N}((0.30, 0.65, 0.50), 0.05^2 I)$, $\varphi_1 = \frac{1}{3}\sum_{n=1}^{3} \mathcal{N}(b_n, 0.04^2 I)$ with $b_1 = (0.70, 0.30, 0.35)$, $b_2 = (0.70, 0.70, 0.35)$, $b_3 = (0.70, 0.50, 0.70)$ (`example_split_3d`). The branch-mass matrix is

$$\omega = \begin{pmatrix} 0.391 & 0.006 & 0.103 \\ 0.006 & 0.391 & 0.103 \end{pmatrix} \quad \text{(rows: source blobs; columns: targets } b_1, b_2, b_3\text{)},$$

i.e. each source sends 78% of its mass to the nearest target, 21% up to the shared target b_3, and 1% across to the far one. ρ_0 has six Gaussian components but only two visible blobs (std 0.045, centres $x \approx 0.37$); ρ_1 has three visible blobs (std 0.038, $x \approx 0.655$). Mass inside the box $\geq 1 - 2 \cdot 10^{-8}$; kinetic energy 0.0640. The 1% cross branch is a good stress test: it is a low-density bridge between two far blobs, exactly where a positivity-challenged scheme or an under-converged LADMM goes wrong first (Fig. 2).

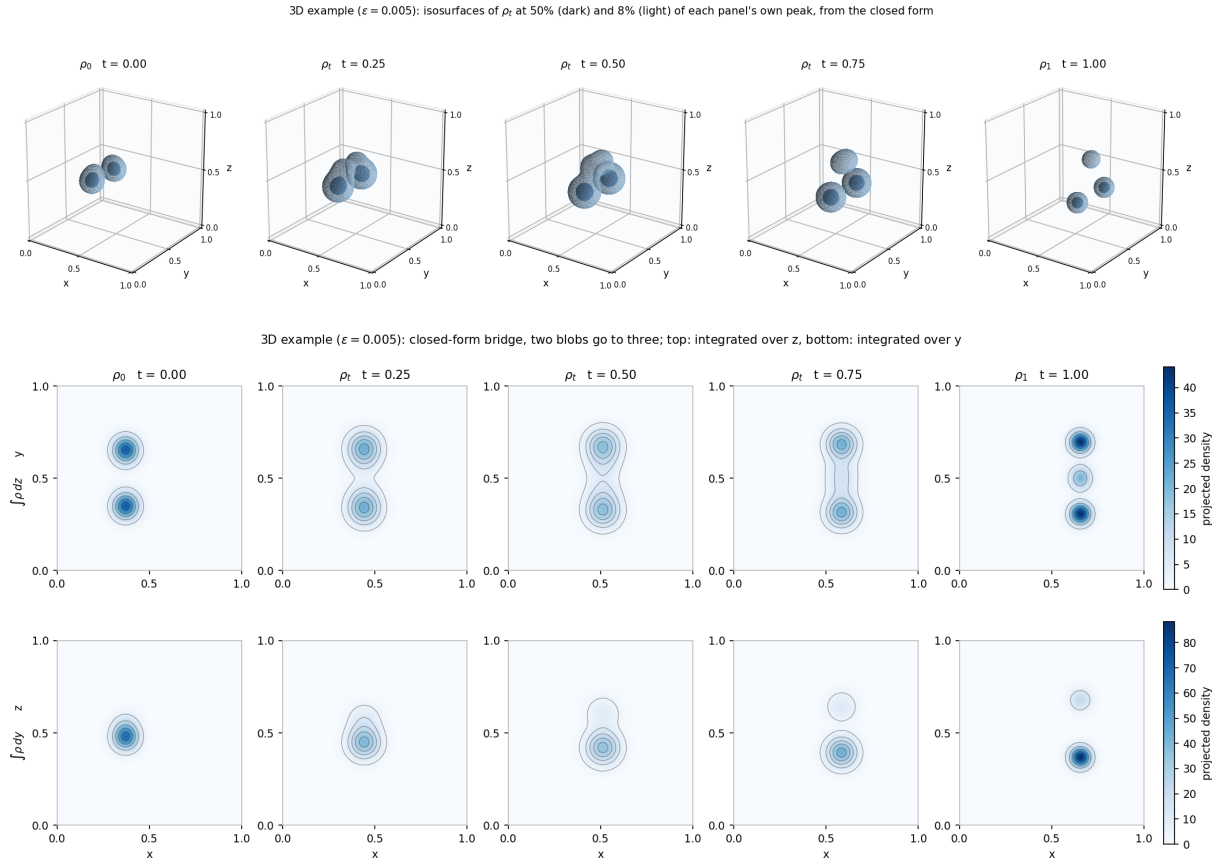


Figure 2: 3D example from the closed form. Top row: isosurfaces of ρ_t at 50% (dark) and 8% (light) of each panel's own peak. Lower rows: projections of ρ_t onto the x–y plane and the x–z plane. Two blobs become three; the third target at $z = 0.70$ receives 21% of the mass and is visible from $t = 0.5$ in the bottom row, while the faint x–y band between the two main blobs at $t = 0.75$ is the 1% cross branch plus the shared branch.

Worked 2D example along the paper's route: a fixed, non-mixture ρ_0. The closed-form variant above only produces Gaussian-mixture marginals. To see whether the construction transfers to a genuinely different ρ_0, the paper's own route was followed, with one numerical step. Code: `python/shared/paper_route_sb.py`, example `example_ring_to_two_2d`. The six objects are obtained in this order (Fig. 4):

1. *Choose* $\varphi_1(y) = \frac{1}{2}\mathcal{N}(y|(0.70, 0.30), 0.04^2 I) + \frac{1}{2}\mathcal{N}(y|(0.70, 0.70), 0.04^2 I)$. This is the paper's LSE-quadratic potential $g^* = \sigma^2 \log \varphi_1$ with $A_n \succ 0$.

2. *Closed form:* $\varphi(t,x) = \sum_n w_n \mathcal{N}(x|b_n, Q_n + 2\varepsilon(1-t)I)$, the backward heat flow. At $t=0$ this is φ_1 blurred by variance 2ε, a smooth two-lobed modulation; it is also $Z(x)$ in (5) up to a constant.

3. *Fix* ρ_0: a smooth ring, $\rho_0 \propto \exp\bigl(-(|x-c|-r)^2/2w^2\bigr)$ with $c=(0.35,0.50)$, $r=0.12$, $w=0.025$, normalized on the grid. It is not a Gaussian mixture.

4. *Forced:* $\hat{\varphi}_0 = \rho_0/\varphi(0,\cdot)$. There is no freedom here; this is what makes the x-marginal come out as ρ_0. In the paper this factor is never written down because the sampler never needs it. Here it is a crescent: the part of the ring facing away from the targets, where $\varphi(0,\cdot)$ is smallest.

5. *The one numerical step:* $\hat{\varphi}(t,\cdot) = \mathcal{N}(0, 2\varepsilon t I) * \hat{\varphi}_0$, the forward heat flow. Since the target points form a tensor grid and the kernel is separable, this is $K_x F K_y^\top$ with F the values of $\hat{\varphi}_0$ on a fine $n_f \times n_f$ grid and K_x the $n \times n_f$ matrix of 1D Gaussian weights times the cell width. Midpoint quadrature of a smooth Gaussian integrand is spectrally accurate: against $n_f = 1024$, the $n_f = 128$ result differs by $1.4 \cdot 10^{-8}$ (relative to the peak) at the smallest time the solver needs, $t = 1/32$, and $n_f = 512$ by $2 \cdot 10^{-10}$; at $t \geq \frac{1}{2}$ all are below 10^{-9}.

6. *Products:* $\rho_t = \varphi(t,\cdot)\,\hat{\varphi}(t,\cdot)$ (no constant, since $\int \rho_0 = 1$), $\rho_1 = \varphi_1 \hat{\varphi}(1,\cdot)$, and $v = 2\varepsilon \nabla \log \varphi(t,\cdot)$ from step 2, closed form and independent of ρ_0; $m = \rho_t v$.

Checks, as for the mixture case: $\int_{[0,1]^2} \rho_t = 1$ to $8 \cdot 10^{-7}$ at all t (the deficit is the mass that diffuses past the walls on $\mathbb{R}^2$ and comes back); ρ_1 equals the paper's definition $\int \rho_0(x)\pi^*(y|x)\,dx$ by direct quadrature to $3 \cdot 10^{-16}$; the finite-difference residual of $\partial_t \rho + \nabla \cdot (\rho v) - \varepsilon \Delta \rho$ is $1.9 \cdot 10^{-3}, 4.7 \cdot 10^{-4}, 1.2 \cdot 10^{-4}$ on $200^2, 400^2, 800^2$ grids at $t = \frac{1}{2}$, i.e. $O(h^2)$, and likewise at $t = 0.1$ and 0.9. Kinetic energy 0.0911. Peak density 21 at $t=0$, 10 at $t=\frac{1}{4}$, 52 at $t=1$. Building the full staggered-grid reference at 32^3 takes 0.3 s.

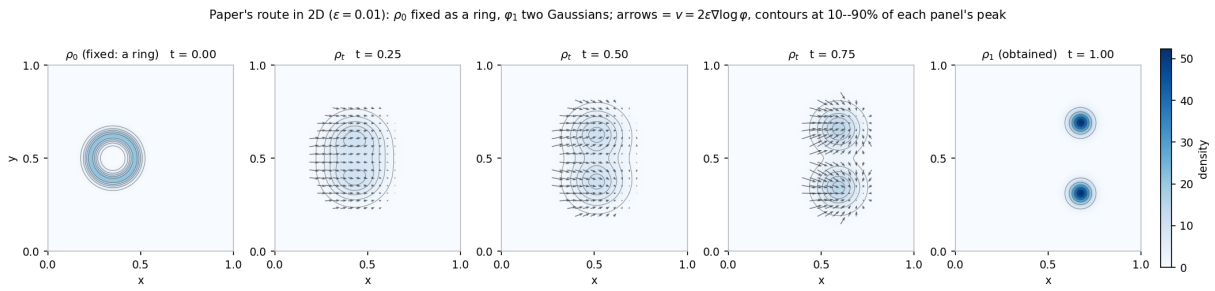


Figure 3: The paper's route in 2D: ρ_0 fixed as a ring, φ_1 two Gaussians. Everything else is derived. The ring fills in and stretches toward the targets by $t = \frac{1}{4}$, is a two-lobed cloud at $t = \frac{1}{2}$, and arrives as two Gaussian-looking blobs. Arrows: $v = 2\varepsilon \nabla \log \varphi$. Shared colour scale, contours at 10–90% of each panel's own peak.

Does this transfer to verifying the grid solver? Yes, with the following accounting. The solver returns ρ on cell centres at $k\,dt$ and m on faces at $(k - \frac{1}{2})dt$; `staggered_reference` evaluates steps 2, 5 and 6 at exactly those points, so the comparison is pointwise, with no interpolation of the reference. The reference error budget is: quadrature in step 5, $\leq 10^{-9}$ for $n_f \geq 256$; the $\mathbb{R}^2$-versus-box mismatch, bounded by the tail mass ($8 \cdot 10^{-7}$ here, set by how far the support sits from the walls); and nothing else, since steps 1–4 and 6 are closed form. Both

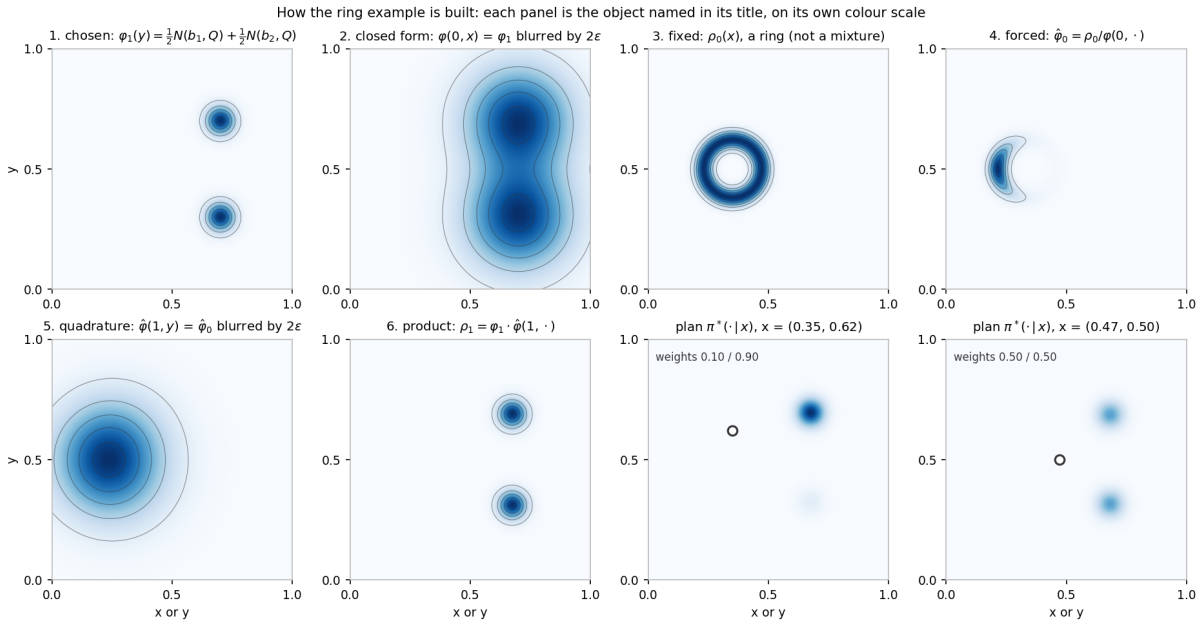


Figure 4: How the ring example is built, numbered as in the text. Top: the chosen factor φ_1, its closed-form blur $\varphi(0, \cdot)$, the fixed ρ_0, and the forced factor $\hat{\varphi}_0 = \rho_0/\varphi(0, \cdot)$, a crescent on the side of the ring away from the targets. Bottom: the one quadrature, $\hat{\varphi}(1, \cdot)$; the product ρ_1; and the static plan $\pi^*(y|x)$ for two source points on the ring, each a two-Gaussian mixture in y whose weights depend on where x sits on the ring (0.10/0.90 at the top of the ring, 0.50/0.50 on the symmetry line).

are far below the $O(h^2)$ discretization error one is measuring (10^{-2}–10^{-3} for $n = 32$–128). In 3D step 5 becomes three tensor contractions, $\hat{\varphi}(t) = F \times_1 K_x \times_2 K_y \times_3 K_z$, with F on a 256^3 grid; cost $O(n\, n_f^3)$ per time slice, seconds on a GPU, so a 96^4 reference is cheap. What one gives up relative to the mixture case is the readable branch-mass table and the closed-form ρ_1; what one gains is complete freedom in ρ_0 (any smooth positive density that is small at the walls) while v stays closed form. Two practical rules. Keep ρ_0 smooth on the scale of $\sqrt{2\varepsilon\, dt}$, the blur at the first time step, or the forced factor $\hat{\varphi}_0$ will be rough. And choose φ_1 so that $\varphi(0, \cdot)$ does not vary wildly across the support of ρ_0, or $\hat{\varphi}_0$ concentrates on a sliver of it: here $\varphi(0, \cdot)$ varies by a factor 261 over the ring, which already turns the ring into the crescent of Fig. 4; a larger ε or targets closer to ρ_0 reduce this. If the wall truncation ever matters, replace the Gaussian kernels in steps 2 and 5 by the Neumann heat kernel of the box (a cosine series, or the method of images); then the reference satisfies the no-flux condition exactly.

Caveats and knobs.

- The paper allows A_n with eigenvalues in $(-1, 0)$ (a locally *repulsive* potential). Those are not Gaussian mixtures and are excluded here; the price is that ρ_1's modes are always blobs around the b_n.
- Blob positions are pulled toward the other side: $c_{in}(0)$ lies between a_i and b_n, more so for larger ε or wider P_i. Keep every $c_{in}(t)$ at least 4–5 standard deviations of $C_{in}(t)$ inside the box; $C_{in}(t)$ is largest near the middle of $[0, 1]$ in t.
- The mass split ω_{in} is an output. To force a given split, tune u_i, w_n (they enter ω linearly) rather than the geometry.
- The same object with $I = N = 1$ and unequal $P \neq Q$ is an unequal-covariance Gaussian bridge, an independent check on Sec. 2.1.

Effort: low (the closed form is written down; no solver runs yet). Paper value: good; multimodal exact references are novel in the grid-solver literature.

2.5 Reflected SB theory (matches the no-flux walls)

Caluya and Halder, "Reflected Schrödinger Bridge: Density Control with Path Constraints", ACC 2021, arXiv:2003.13895: SB with reflecting boundaries. The forward–backward PDE system has a no-flux Neumann condition on the forward factor and an oblique Neumann condition on the backward factor. Useful to cite as the exact continuous problem the discretization targets, and to justify tests where mass does touch the walls (e.g. uniform → bump inside the box).

3 Application problems

3.1 Cosmology: Lagrangian reconstruction (MAK) as dynamic OT

Strongest narrative, with a boundary-condition mismatch to handle.

Literature. Frisch, Matarrese, Mohayaee, Sobolevski, Nature 417:260 (2002); Brenier, Frisch, Hénon, Loeper, Matarrese, Mohayaee, Sobolevskii, "Reconstruction of the early Universe as a convex optimization problem", MNRAS 346:501 (2003) (MAK; uniqueness for convex bounded

domains); Lévy, Mohayaee, von Hausegger, MNRAS 506:1165 (2021), fast semi-discrete OT; von Hausegger, Lévy, Mohayaee, PRL 128:201302 (2022), BAO via semi-discrete OT; Nikakhtar, Sheth, Lévy, Mohayaee, PRL 129 (2022), OT reconstruction of BAO; Nikakhtar, Padmanabhan, Lévy, Sheth, Mohayaee, arXiv:2307.03671 (redshift space); arXiv:2606.01529 (primordial non-Gaussianity). The problem is exactly displacement interpolation from a uniform initial density to the $z = 0$ density (Zel'dovich approximation $\Leftrightarrow$ quadratic-cost OT).

Schrödinger-bridge version. The survey found *no* paper doing an entropic or SB cosmological reconstruction (searches for "Schrödinger bridge/problem" with cosmology return only the unrelated Schrödinger–Poisson method and Bunne-style SB theory). A 3D SB reconstruction with ε read as a small velocity-dispersion / multistreaming smoothing would be a genuinely new, if modest, contribution; the physical meaning is heuristic, so say so rather than claim it.

Boundary conditions. All simulation tests use a periodic box (Brenier et al. 2003; Lévy et al. 2021; Nikakhtar et al.). Observational applications are not periodic: Lavaux, Mohayaee, Colombi, Tully, Bernardeau, Silk, MNRAS 2008, pad a $100\,h^{-1}$Mpc-deep 2MRS volume with a homogeneous particle distribution and trust only the inner $60\,h^{-1}$Mpc. Practical recipe for our box: take a sub-cube, pad by a margin at least the typical displacement (about $10\,h^{-1}$Mpc at $z = 0$), and evaluate only the interior; no-flux is then a controlled approximation. Alternatively multiply by a smooth window. Flag this in the paper.

Data. CAMELS Multifield Dataset (CMD), Villaescusa-Navarro et al., ApJS 259:61 (2022): 3D grids of $M_{\rm tot}$, $M_{\rm cdm}$, $M_{\rm gas}$, etc. at 128^3, 256^3, 512^3 voxels over $(25\,h^{-1}{\rm Mpc})^3$ at redshifts $0, 0.5, 1, 1.5, 2$; float32 `.npy`, 8 MB per 128^3 grid; periodic; https://camels-multifield-dataset.readthedocs.io (Globus / binder / URL). Quijote (Villaescusa-Navarro et al. 2020): 3D density fields at 64–512^3 over $(1000\,h^{-1}{\rm Mpc})^3$, CIC/PCS, `.npy`, https://quijote-simulations.readthedocs.io/en/latest/df.html. A 128^3 CMD $z = 2 \to 0$ pair downsampled to 96^3 (or $z = 1 \to 0$ to keep displacements small) is a ready-made problem. Uniform $\to z = 0$, the MAK problem proper, also works: the initial density is uniform by construction.

What ρ_0 and ρ_1 are on a grid. Brenier et al. discretize into N equal point masses because their solver is an assignment algorithm: the N particles of an N-body simulation at $z = 0$ are matched to the N sites of a regular Lagrangian lattice, which is the uniform initial state. The grid version of the same two measures is:

- $\rho_0(q) \equiv 1$ (in units of the mean density) on the Lagrangian grid $q \in [0, L]^3$. At decoupling the fluctuations are $\delta \sim 10^{-5}$, so "uniform" is exact for every purpose here. This is the continuum limit of Brenier's lattice of equal masses; the lattice spacing is what they call the mesh ($6.25\,h^{-1}$Mpc in their tests) and plays the role of our cell size.
- $\rho_1(x)$ on the Eulerian grid $x \in [0, L]^3$: the $z = 0$ mass deposited on the cells by cloud-in-cell (CIC) assignment, divided by the mean, so that $\int \rho_1 = \int \rho_0$ automatically. The public density fields (Quijote, CAMELS) are exactly this object; from a particle snapshot it is one `np.add.at`. A floor of a few percent of the mean for empty voids is the only preprocessing.

The dimension is three: the physical problem is 3D, the data are 3D, and the reconstruction literature is 3D. A 2D version is legitimate for development only if it is generated as a 2D flow (a

2D potential and 2D displacement), because a 2D slice of a 3D flow does not conserve mass within the slice. The quadratic cost in comoving coordinates and growth-factor time $t = D(z)/D(0)$ are the same in any dimension.

The problem as the solver sees it (Fig. 5). Fig. 5 shows a realization built without any download (`python/shared/zeldovich.py`): a Gaussian random linear density field with a BBKS ΛCDM-like spectrum ($\Gamma = 0.21$, $n_s = 0.96$, $\sigma_8 = 0.8$) is synthesized in the cosine basis of a $512\,h^{-1}$Mpc box, so the displacement potential is even about every wall and the Zel'dovich displacement $\Psi = -\nabla\Phi$, $\nabla^2\Phi = \delta$, has zero normal component on the walls. 128^3 particles are placed on the Lagrangian grid and moved to $x = q + D\,\Psi(q)$, and ρ_1 is their CIC deposit on 96^3 cells of $5.3\,h^{-1}$Mpc, the resolution the solver can afford. Numbers for this realization at $D = 1$ ($z = 0$): rms displacement $9.5\,h^{-1}$Mpc, i.e. 1.8 cells, maximum $33\,h^{-1}$Mpc; ρ_1 ranges from empty cells to 29× the mean, with 2.2% of cells below 0.1 and 1.1% above 5; 15% of Lagrangian cells are shell-crossed ($\det(I + \nabla\Psi) \le 0$). At $D = 0.5$ ($z \approx 1$) the range is 0.05–8.7 with 0.07% shell-crossed cells, and at $D = 0.3$ ($z \approx 2$) it is 0.2–4.7 with none.

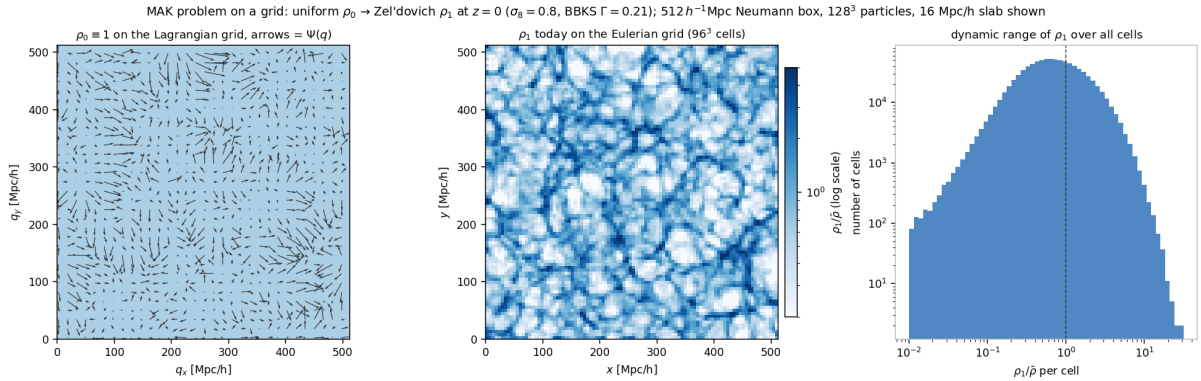


Figure 5: The MAK problem on the solver's grid. Left: $\rho_0 \equiv 1$ on the Lagrangian grid, with the Zel'dovich displacement $\Psi(q)$ of a $16\,h^{-1}$Mpc slab drawn as arrows; this field is the *answer* the reconstruction has to recover. Middle: ρ_1, the same slab today, on 96^3 cells, log colour scale: the arrows converge onto the filaments and knots and diverge out of the voids. Right: the histogram of ρ_1 over all cells, the dynamic range the prox has to handle.

What the figure means physically. Every arrow on the left is the comoving path of a parcel of dark matter from the nearly uniform early Universe to where it sits today; in the Zel'dovich approximation the parcels move in straight lines at constant comoving velocity, with the growth factor D as the clock, and Ψ is a gradient field. Where arrows converge, sheets form first ("pancakes"), then filaments where sheets intersect, then the knots that host clusters; where they diverge, voids empty out. The reconstruction problem inverts the middle panel into the left one: given only today's density, find the gradient displacement that produced it. Brenier's theorem says this inverse is unique as long as the map is the gradient of a convex potential, i.e. before shell crossing, and that it is the quadratic-cost optimal transport from ρ_0 to ρ_1. In solver terms: $\rho(t,\cdot)$ at $t = D(z)/D(0)$ is the density at redshift z, and the velocity m/ρ at $t \to 0$, read on the uniform grid, *is* the reconstructed $\Psi(q)$, which is also the peculiar velocity field up to the factor fHa. The 15% shell-crossed cells at $z = 0$ are where multistreaming has erased the inner

velocity structure; this is the one place where a small ε has a plausible reading, as a velocity dispersion, and the reason the exact-answer test should be run at $D \approx 0.3$–0.5 where no crossing has occurred and the arrows in the left panel are the exact solution.

Sanity check. Compare the OT displacement field against the true Lagrangian displacements available from particle IDs (needs the raw snapshots, on the CAMELS Globus), or at least against the MAK / semi-discrete solution (Lévy's code is public). The dynamic range at $z = 0$ is huge (voids ~ 0.1, filaments $> 100\times$ the mean): expect stiffness, and consider a smoothing scale as Nikakhtar et al. do.

Effort: moderate. Paper value: high; a recognizable application with a large literature and a periodic-versus-no-flux discussion reviewers will find honest.

3.2 Cryo-EM density maps (EMDB)

The best real-data fit to the box: maps are already on uniform Cartesian grids with zero density at the box edges (ideal for no-flux), nonnegative after thresholding, mass renormalizable; a conformational change is a physically meaningful displacement interpolation.

Literature doing this with static entropic OT. Ecoffet, Poitevin, Dao Duc, "MorphOT: transport-based interpolation between EM maps with UCSF ChimeraX", Bioinformatics 36:5528 (2020), bioRxiv 10.1101/2020.09.08.286302; Ecoffet et al., "Application of transport-based metric for continuous interpolation between cryo-EM density maps", AIMS Mathematics 7(1) (2022), PMC9377683: entropic W_2 barycenters via Gaussian convolution (Solomon et al.), tested at 60^3, 120^3, 240^3, default downsampling to 60^3, $\gamma = 1$, at most 1500 iterations; they assume equal mass and list unbalanced OT as future work. Alignment: Riahi, Woollard, Poitevin, Condon, Dao Duc, "AlignOT", IEEE TCBB 2023 (arXiv:2210.09361); Riahi et al., "EMPOT", arXiv:2311.00850; PRX Life 3:023003 (2025). Zelesko, Moscovich, Kileel, Singer, ISBI 2020, "Earthmover-based manifold learning", used a simulated ATP-synthase rotary dataset (public availability not verified).

Concrete pairs. Used by MorphOT: the Mm-cpn group II chaperonin, EMD-5137 (wild-type closed, 4.3 Å) and EMD-5139 (wild-type open, 8 Å), Zhang et al., Nature 463:379 (2010); also the lidless EMD-5138 / EMD-5140. Same particle, same lab, same box: align rigidly (AlignOT / ChimeraX) and resample to 64–96^3. Alternatively the E. coli ATP synthase rotational states EMD-20006 / EMD-20008 (Sobti et al., eLife 2019), three rotary states of the same complex.

Reference. MorphOT's Sinkhorn barycenters at the same γ (they use an $\mathbb{R}^3$ Gaussian kernel, not Neumann; the difference should be negligible with zero-padded maps), plus our own DCT-Sinkhorn (Section 2.3). Our dynamic solver adds what they lack: momentum / velocity fields (which map to residue motions) and an ε that plays the role of their blur.

Effort: low to moderate (download from `https://www.ebi.ac.uk/emdb/`, threshold, align, resample). Paper value: high; "static barycenter vs. dynamic bridge on EMD-5137 $\to$ 5139" is a clean figure.

Can it be run? Yes, 3D, and it is the cleanest real-data fit. An EMDB entry is a single-precision 3D array on a cubic voxel grid (typically 1–4 Å per voxel, 200–400 voxels per

side) with the particle centred and solvent at the edges. ρ_0 and ρ_1 are two maps of the same complex in two conformations: threshold at zero (map noise is signed), optionally mask the solvent, align rigidly, resample both to 96^3, and normalize. Equal mass is physically exact, since it is the same molecule, and zero density at the walls is automatic. Fig. 6 shows a phantom with the architecture of the MorphOT pair EMD-5137/5139: two stacked eight-fold rings of subunits, each with an apical "lid" domain that folds over the central cavity in the closed state and swings outward in the open state. Interpretation: ρ_t is the conformational transition path between the two states, the quantity MorphOT produces as a static barycenter sequence, and m/ρ is the domain motion field, which can be read against the atomic models fitted to each map (the lid domains should move outward and up, the equatorial ring should barely move). ε **has no physical meaning as a diffusion**; its honest reading is the resolution of the maps: a map at 8 Å is a sharper map blurred by a kernel of that width, so matching $\sqrt{2\varepsilon}$ to the lower of the two resolutions (here 8 Å, about 1–2 cells at 96^3) makes the bridge's blur commensurate with the data's, and MorphOT's γ is the same blur in their convention. The phantom's density spans zero to 42× the mean inside the particle envelope, so a floor in the solvent region is the only stiffness precaution.

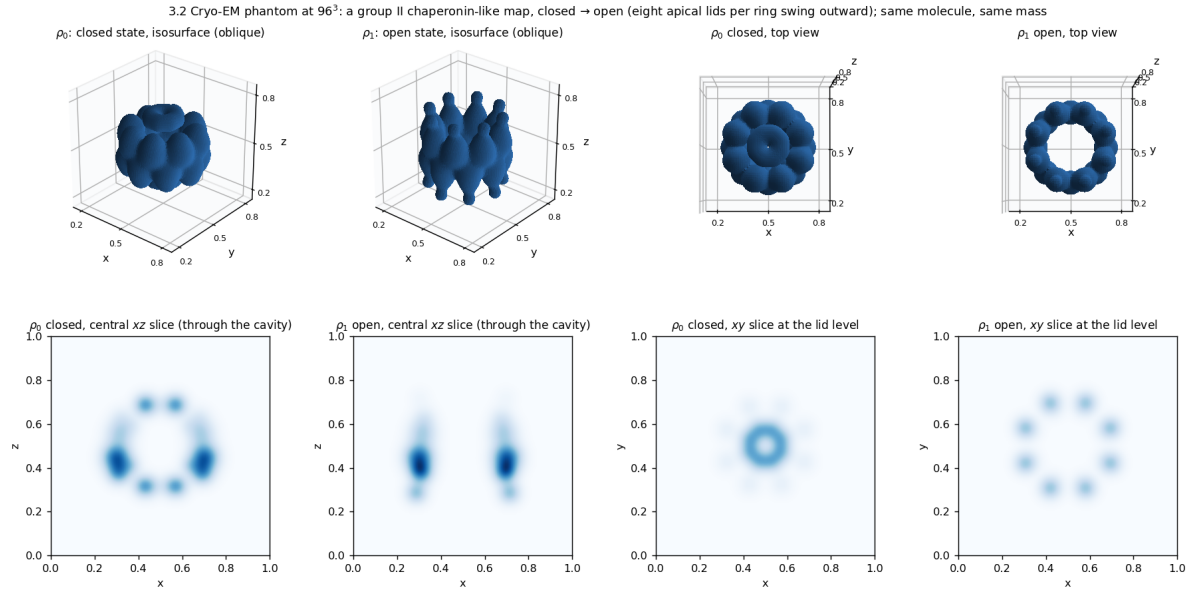


Figure 6: Cryo-EM phantom at 96^3. Top: isosurfaces at 25% of the peak for the closed (ρ_0) and open (ρ_1) states, oblique and from the top; the eight lids of the upper ring close the cavity in ρ_0 and open into a flower in ρ_1. Bottom: a central vertical slice through the cavity and a horizontal slice at the lid level, same colour scale.

Smaller alternative: molecular electron densities. Gaussian `.cube` files are uniform grids (e.g. EDBench, arXiv:2505.09262, about 3.4M molecules with cube files; QM9-derived densities at 256^3 / 0.1 bohr). A chemRxiv 2025 preprint, "Electron Density Transport During Chemical Reactions" (10.26434/chemrxiv, April 2025), already uses POT / Sinkhorn on cube densities along reaction paths. Two snapshots of a reaction (reactant and product densities) have equal electron count, so equal mass automatically: a tidy small problem. Effort: low. Paper value:

moderate, novel.

3.3 Medical imaging

OT in 3D registration. Haker, Zhu, Tannenbaum, Angenent, IJCV 60:225 (2004), Monge–Kantorovich warping of 3D brain MRI. Chen, Elkin, Tannenbaum et al., "Visualizing fluid flows via regularized optimal mass transport with applications to neuroscience", J. Sci. Comput. (2023), and "Unbalanced regularized optimal mass transport with applications to fluid flows in the brain", Sci. Rep. (2024), https://www.nature.com/articles/s41598-023-50874-y: a Benamou–Brenier plus diffusion (advection–diffusion) model, essentially our PDE, solved on cell-centred grids of 50^3 (synthetic) and $56 \times 106 \times 51$ (rat-brain DCE-MRI, glymphatic tracer). This is the closest existing 3D application of our exact formulation, and there the diffusion coefficient has physical meaning (tracer diffusivity). The rat data do not appear to be public, but the model and grid sizes are a direct benchmark to cite.

Datasets.

- DIR-Lab 4DCT (https://www.dir-lab.com/): 10 cases, 10 respiratory phases, about $1 \times 1 \times 2.5$ mm, 300 landmark pairs between end-exhale and end-inhale, which give a sanity check on the transport map.
- Learn2Reg OASIS brain MRI (https://learn2reg.grand-challenge.org/): 416 skull-stripped 1 mm T1 scans cropped to $160 \times 224 \times 192$, with label maps for Dice checks.
- BrainWeb phantoms (https://brainweb.bic.mni.mcgill.ca/): $181 \times 217 \times 181$ at 1 mm with ground-truth tissue classes.
- ACDC cardiac cine (Bernard et al., IEEE TMI 2018): in-plane 1.4–1.7 mm but slices 5–8 mm; anisotropic, a poor fit for a cubic grid.

Can it be run? Yes, 3D. Both sub-problems are scalar fields on a cubic voxel grid, so ρ_0 and ρ_1 are read off directly after masking, resampling to 96^3 and renormalizing. Fig. 7 shows synthetic stand-ins with the geometry of the real data (no download was possible here).

- *Glymphatic DCE-MRI (top row).* ρ_0 = tracer concentration map shortly after injection into the cisterna magna (posterior, ventral); ρ_1 = the same map about an hour later, when tracer has run along the perivascular spaces of the major arteries and begun to enter the parenchyma. Interpretation of the solution: m/ρ is the bulk flow of cerebrospinal fluid along the perivascular channels, which is what Chen et al. extract and plot as pathlines; ρ_t interpolates the missing frames. ε **has a physical meaning**: it is the tracer's effective diffusivity in tissue, and their rOMT model is literally our constraint. In their units $\sigma^2 \approx 0.002$ per voxel^2 per frame, about $3 \cdot 10^{-6}$ on the unit box, i.e. a regularization-sized value rather than a resolved diffusion. Clearance makes the real problem unbalanced; equal mass is an approximation to state.
- *Lung 4DCT (bottom row).* ρ_0 = tissue density at end-exhale (CT number converted to $1 -$ air fraction), ρ_1 = at end-inhale: the lobes lengthen along the head–feet axis as the diaphragm descends and the density drops accordingly, with the posterior (dependent) lung denser. Tissue mass is conserved by breathing, so equal mass is physically right for this choice of ρ. Interpretation: m/ρ is the breathing motion field, checked against the 300 DIR-Lab

landmark pairs. ε **has no physical meaning** (tissue does not diffuse); it only regularizes the registration.

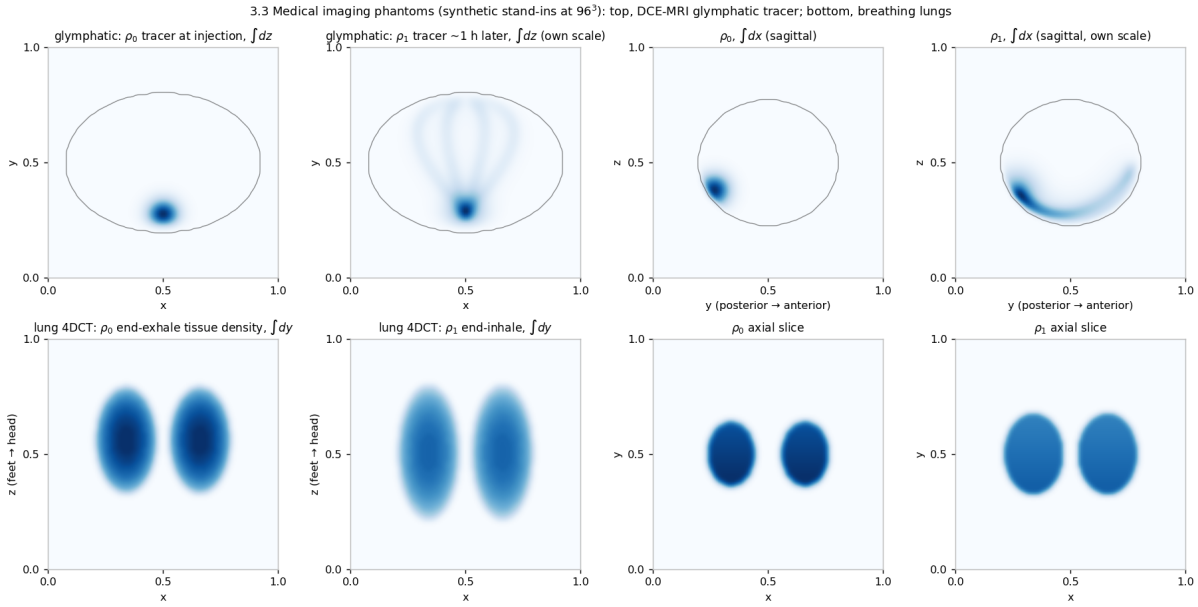


Figure 7: Medical-imaging phantoms at 96^3. Top: glymphatic tracer, injected posteriorly and ventrally (ρ_0), then spread along four perivascular channels on the ventral and lateral brain surface with diffuse parenchymal uptake (ρ_1, drawn on its own colour scale because its peak is $10\times$ lower); the outline is the brain mask. Bottom: two lung lobes at end-exhale and end-inhale, projected along the anterior–posterior axis and as an axial slice; the inhale lobes are 25% longer in z and correspondingly less dense.

Caveats. Intensities are not conserved mass: lung CT density genuinely changes with inflation, so mass is not equal; brain MRI intensity is only roughly proportional to tissue mass. One must renormalize and say so. DIR-Lab lungs are the most "mass-like" (HU tracks tissue density) and have real motion. Skulls and the table touch the box edges unless masked. Effort: moderate. Paper value: moderate; expected, and reviewers know unbalanced OT is the honest model here (cite the 2024 unbalanced rOMT).

3.4 Fluids and geoscience

- JHTDB (https://turbulence.pha.jhu.edu/), "Homogeneous buoyancy driven turbulence": 1024^3 periodic, 1015 frames at $\Delta t = 0.04$, stores density (Atwood 0.05, Schmidt 1); cutouts via the web / SciServer API. Two density frames on a 96^3 sub-cube give a "transport a mixing scalar" problem. But the field is periodic and the physical evolution is advection–diffusion by a turbulent velocity, not OT, so the check is qualitative (OT velocity vs. DNS velocity at large scales). Here ε has physical meaning (molecular diffusivity $\kappa = \nu/\mathrm{Sc}$) only if the domain matches. A dedicated passive-scalar isotropic dataset could not be confirmed.
- ERA5 pressure levels (Copernicus CDS, https://cds.climate.copernicus.eu/datasets/reanalysis-era5-pressure-levels): specific humidity and ozone mass mixing ratio on 37

levels, 0.25°, hourly. The lon-lat-pressure cube is physically non-uniform (air-mass weighting, log-pressure vertical) and horizontally periodic; workable for a regional box with weak boundary fluxes (a Saharan dust or ozone event), but the metric is not Euclidean. Fit: weak, mostly illustrative.

- OT data assimilation: Feyeux, Vidard, Nodet, NPG 25:55 (2018), 2D shallow water, https://npg.copernicus.org/articles/25/55/2018/; Bocquet, Vanderbecken, Farchi, Dumont Le Brazidec, Roustan, NPG 31:335 (2024), "Bridging classical data assimilation and optimal transport: the 3D-Var case", https://npg.copernicus.org/articles/31/335/2024/; Farchi, Bocquet, Roustan, Mathieu, Quérel, Tellus B 68:31682 (2016), Wasserstein comparison of Fukushima radionuclide plumes (public solver at http://cerea.enpc.fr/opttrans/). These are 2D; a 3D plume (FLEXPART / Polyphemus output) would be new but requires a model run. Effort: high. Paper value: moderate.

Can it be run? Yes, 3D for two of the three; the third is 2D. Fig. 8 shows the two 3D stand-ins.

- *Atmospheric dispersion (top row).* ρ_0 = concentration of a pollutant puff just after release, ρ_1 = the same puff after time T, carried downwind and spread by turbulent eddy diffusivity K. This is the 3D version of the Farchi / Bocquet plume comparisons, which were done on column-integrated 2D maps. ε **has an exact physical meaning**: with $\varepsilon = K$ (in box units) the physical evolution is advection–diffusion with a uniform wind, the reference process itself has that law, so the SB between ρ_0 and ρ_1 *is* the true plume at every t and m/ρ is the wind. This makes the puff a verification problem with a physical ε, and it degrades gracefully: a sheared wind gives a plume the SB can only approximate, which is the honest version of the data-assimilation use. Real plumes have $K_h \gg K_v$; our isotropic ε is the idealization.
- *Buoyancy-driven mixing (bottom row, JHTDB stand-in).* ρ_0 = heavy fluid fraction with a nearly flat interface, ρ_1 = the same scalar after plumes have formed. Griddable at 96^3 from a JHTDB cutout, but the physical velocity is divergence-free and turbulent, not a gradient, so the OT velocity is at best a large-scale proxy; and a cutout exchanges mass through its faces, so renormalization is forced. ε **= molecular diffusivity** κ is physical only for the full periodic box. Fit: weak.
- *ERA5 humidity / ozone.* 3D (lon, lat, pressure) but on a non-Euclidean, air-mass-weighted grid; only a small regional box is a reasonable approximation. Illustrative at best; not plotted.

3.5 Shape interpolation (computer graphics)

Solomon, de Goes, Peyré, Cuturi, Butscher, Nguyen, Du, Guibas, "Convolutional Wasserstein distances", SIGGRAPH 2015: 3D indicator-function interpolation and barycenters on 60^3 and 100^3 voxel grids, code https://github.com/gpeyre/2015-SIGGRAPH-convolutional-ot. Papadakis, Peyré, Oudet, SIIMS 7:212 (2014), proximal-splitting Benamou–Brenier with obstacles; released MATLAB code runs 2D+time at $32 \times 32 \times 32$, https://github.com/gpeyre/2013-SIIMS-ot-splitting. Lévy, ESAIM M2AN 49:1693 (2015), semi-discrete L^2 OT in 3D (tetrahedral meshes, 10^5–10^6 Diracs). Haber and Horesh, NMTMA 8:97 (2015), multilevel time-dependent OT (3D sizes not confirmed). Stanford bunny / armadillo / dragon meshes are public (Stanford 3D Scanning Repository) and voxelize trivially; ShapeNet similar.

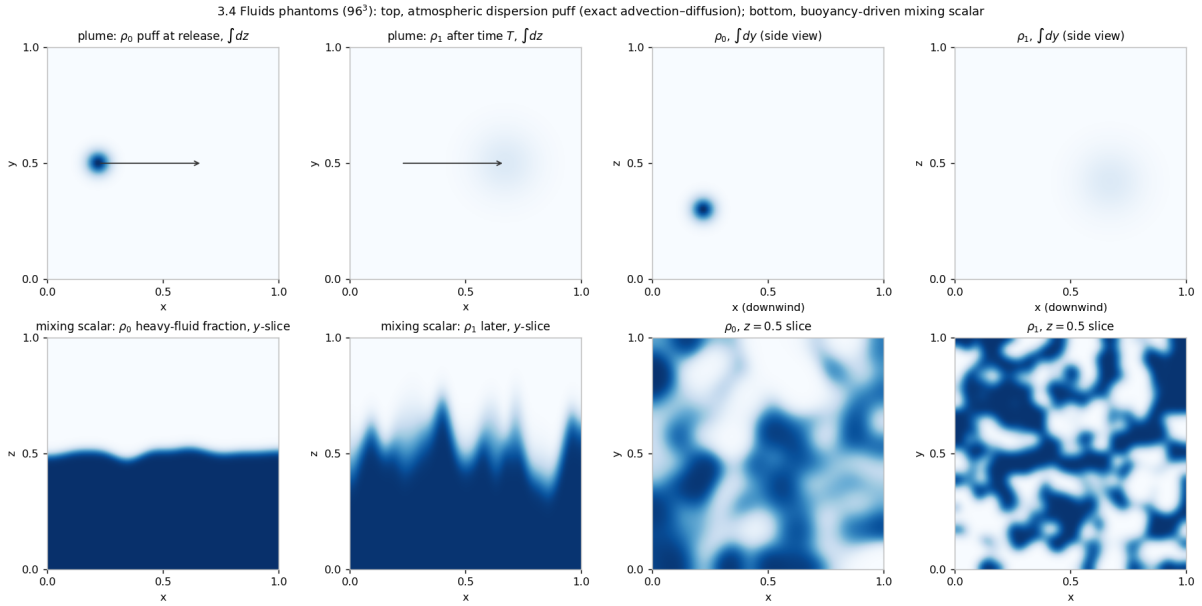


Figure 8: Fluids phantoms at 96^3. Top: a Gaussian puff released at $x = 0.22$ (ρ_0) and the exact advection–diffusion solution after unit time with wind $(0.45, 0, 0.12)$ and $K = 0.004$ (ρ_1), shown integrated along z and along y; the arrow is the wind. Bottom: heavy-fluid fraction before and after mixing, in a vertical and a horizontal slice.

Fit: uniform indicator densities are discontinuous (sharp fronts stress positivity; ε smooths them), zero at the walls, equal mass after normalization. Reference: Solomon's convolutional barycenters at matched blur, plus our DCT-Sinkhorn. Effort: low. Paper value: moderate (familiar to OT reviewers, good figures, shows ε-smoothing of shocks); the 100^3 Solomon grid is a natural resolution comparison point.

Can it be run? Yes, 3D. ρ_0 and ρ_1 are the indicator functions of two voxelized solids, smoothed by one cell and normalized (Fig. 9: a torus and a sphere of equal volume, which also changes topology). Interpretation: ρ_t is the Wasserstein geodesic between the shapes, the "morph" of Solomon et al.; m/ρ is the deformation velocity. ε **has no physical meaning.** It is the entropic blur: Solomon's convolutional kernel is the heat kernel with variance 2ε in our units, so matching their γ to our ε gives a direct comparison, and the figure to make is the sharpening of the front as $\varepsilon \to 0$.

3.6 Single-cell trajectory inference and density control

Trajectory inference. Schiebinger et al., Cell 176:928 (2019), Waddington-OT: 236k cells, 39 time points (h5ad about 1.4 GB via CellRank's `reprogramming_schiebinger`, or https://broadinstitute.github.io/wot/); Lavenant, Zhang, Kim, Schiebinger, "Towards a mathematical theory of trajectory inference" (2021; Ann. Appl. Probab. 2024): SB with the reference diffusion as the model, ε the assumed cell diffusivity. Gridding a 3D PCA / PHATE embedding (KDE at 64^3) of day-0 vs. day-18 cells gives an SB with a physically motivated ε, and growth (unbalanced) issues; no published 3D-gridded version was found. Paper value: moderate, novel; effort moderate; expect reviewers to ask why not the 30-plus intermediate time points

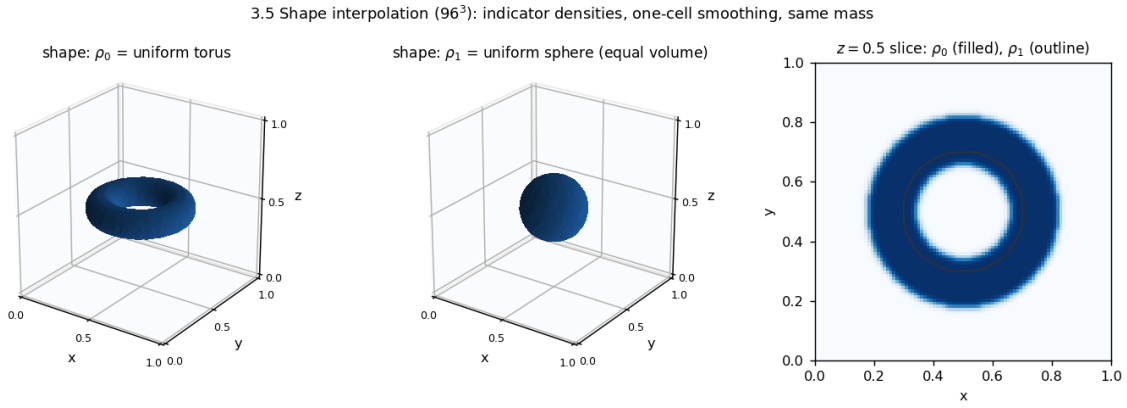


Figure 9: Shape-interpolation phantom at 96^3: a torus ($R = 0.24$, $r = 0.085$) and a sphere of the same volume ($r = 0.20$), uniform inside, one-cell smoothing, same mass; right, the mid-plane slice with the sphere's outline over the torus.

(multi-marginal).

Density control. Chen, Georgiou, Pavon, "Optimal steering of a linear stochastic system to a final probability distribution" I–III, IEEE TAC 2016 / 2018, and Annual Review of Control 2021, "Optimal transport in systems and control": a swarm of Brownian agents steered between two 3D formations (ρ_0 a ring or torus, ρ_1 letters or a cube) with reflecting arena walls is exactly the Caluya–Halder reflected SB; our solver gives the feedback field $v(t, x)$. Synthetic, no data needed. Paper value: moderate (the controls audience likes it).

Can they be run? Yes, both in 3D (Fig. 10).

- *Trajectory inference (top row).* The "space" is a 3D embedding of gene expression (first three PCA or PHATE coordinates, rescaled to the box), so the dimension is three by construction, not by physics. ρ_0 = kernel density estimate of the day-0 cells, one compact cluster; ρ_1 = KDE of the day-18 cells: several clusters, one per fate, with unequal weights, plus a faint bridge of cells still in transit. Interpretation: ρ_t is the inferred population at intermediate days, m/ρ the developmental velocity field, and the plan gives fate probabilities per starting cell. ε **has a model meaning**: Lavenant et al. define trajectory inference as the SB whose reference diffusion is the cells' stochastic gene-expression drift, so ε is the assumed cell diffusivity in embedding coordinates, a parameter to be fitted, not measured. Cell growth and death make the honest model unbalanced.
- *Density control (bottom row).* A swarm of agents with Brownian noise is steered from one formation to another at minimum control energy: ρ_0 = agents on a ring, ρ_1 = agents filling a cube. Reflecting arena walls are exactly our Neumann box (Caluya–Halder). Interpretation: $v = m/\rho$ is the optimal feedback control law each agent applies at (t, x), and ρ_t is the swarm en route. ε **has an exact physical meaning**: $2\varepsilon = \sigma^2$, the intensity of the agents' noise, i.e. how much the formation would smear if uncontrolled. Synthetic by nature, so no data issue at all.

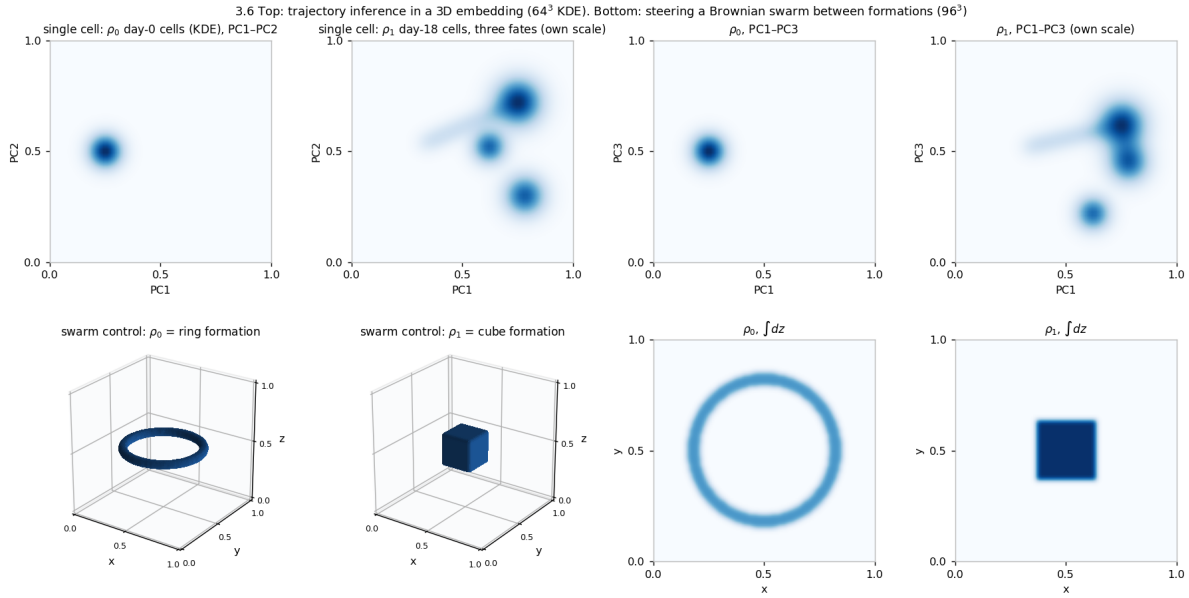


Figure 10: Top: trajectory-inference phantom on a 64^3 embedding grid, day-0 cells and day-18 cells with three fates of weights 0.5/0.3/0.2 and a transit bridge, projected onto two coordinate planes (ρ_1 on its own colour scale). Bottom: swarm control at 96^3, a ring formation and a cube formation, as isosurfaces and integrated along z.

Summary of Secs. 3.2–3.6.

problem	grid	$\rho_0 \to \rho_1$	meaning of ε
cryo-EM conformations	3D	closed → open map	map resolution (blur), MorphOT's γ
glymphatic DCE-MRI	3D	tracer early → late	tracer diffusivity (physical, tiny)
lung 4DCT	3D	tissue density exhale → inhale	none (regularizer)
dispersion puff	3D	release → downwind	eddy diffusivity K (exact)
JHTDB mixing cutout	3D	interface → plumes	molecular κ (full box only)
plume assimilation (Farchi)	2D	column maps	eddy diffusivity (heuristic)
ERA5 ozone / humidity	3D, non-Euclidean	regional box	none
shape morphing	3D	torus → sphere	entropic blur (Solomon's γ)
trajectory inference	3D embedding	day 0 → day 18	cell diffusivity (model parameter)
swarm density control	3D	ring → cube	agent noise $\sigma^2/2$ (exact)

4 Ranked shortlist

1. **Anisotropic / unequal-covariance Gaussians** (Bunne 2023, Janati 2020), including the self-bridge $\rho_0 = \rho_1$: mandatory verification; cost, ρ_t and m all closed form. [verification]

2. **Gushchin-style LSE mixture pairs in 3D** (multimodal, exact drift, ρ_1 by quadrature) plus the product-of-1D bridges; DCT-Sinkhorn as the grid-exact reference for everything else. [verification]

3. **Cryo-EM: EMD-5137 → EMD-5139** (Mm-cpn closed → open) at 64–96^3, compared with MorphOT's entropic barycenters. [application; best box / no-flux fit]

4. **Cosmology: CAMELS CMD** 128^3 M_{tot} **grids**, $z = 1$ (or uniform) $\to z = 0$ on a padded sub-cube, with displacement-field checks against the Lagrangian truth or Lévy's semi-discrete MAK; explicitly discuss periodic vs. no-flux and ε as a heuristic velocity-dispersion smoothing. [application; highest narrative value]
5. **Lung 4DCT** (DIR-Lab, end-exhale $\to$ end-inhale) with landmark-based checks, framed as approximately balanced; cite rOMT (Chen / Tannenbaum 2023–24) as the closest prior 3D advection–diffusion OT. [application]
6. **3D shape interpolation** (Stanford bunny $\to$ armadillo at 64–100^3) replicating Solomon et al. 2015 resolutions, showing ε-smoothing of fronts. [application / benchmark]

Where ε is physical. Glymphatic / tracer rOMT (diffusivity); buoyancy or passive-scalar turbulence (molecular diffusivity, only with matching boundary conditions); single-cell SB (cell-state diffusivity in Lavenant et al.); swarm steering (agent noise). In cosmology and cryo-EM it is a regularization or blur (MorphOT's γ), not a physical constant; no SB-cosmology paper was found, so any "velocity dispersion" reading should be presented as a heuristic.

Key links

- arXiv 2202.05722 (Bunne), 2006.02572 (Janati), 2006.03416 (Mallasto), 2306.10161 (Gushchin; https://github.com/ngushchin/EntropicOTBenchmark), 2003.13895 (Caluya–Halder).
- CAMELS CMD https://camels-multifield-dataset.readthedocs.io; Quijote https://quijote-simulations.readthedocs.io/en/latest/df.html.
- EMDB https://www.ebi.ac.uk/emdb/ (EMD-5137 / 5139; Nature 463:379); MorphOT bioRxiv 10.1101/2020.09.08.286302 and AIMS Math 7(1) 2022 (PMC9377683).
- DIR-Lab https://www.dir-lab.com/; Learn2Reg https://learn2reg.grand-challenge.org/; BrainWeb https://brainweb.bic.mni.mcgill.ca/; rOMT Sci. Rep. https://www.nature.com/articles/s41598-023-50874-y.
- JHTDB https://turbulence.pha.jhu.edu/; ERA5 https://cds.climate.copernicus.eu/datasets/reanalysis-era5-pressure-levels; Bocquet et al. 2024 https://npg.copernicus.org/articles/31/335/2024/; Feyeux et al. 2018 https://npg.copernicus.org/articles/25/55/2018/.
- Solomon et al. code https://github.com/gpeyre/2015-SIGGRAPH-convolutional-ot; WOT https://broadinstitute.github.io/wot/.